

PROJECT DESIGN PHASE

Solution Architecture

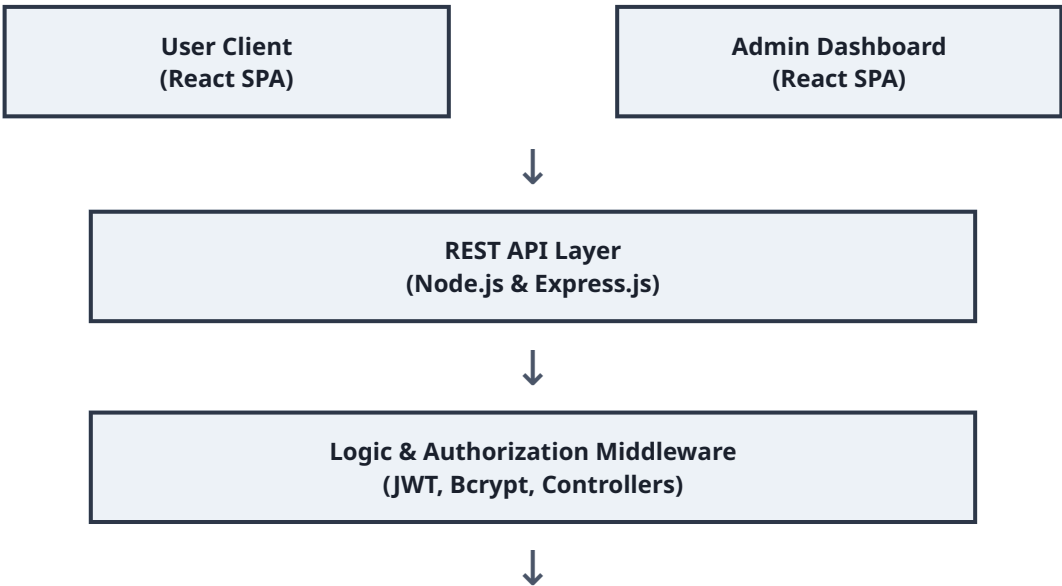
Date	18 June 2026
Team ID	
Project Name	ShopEz (Full Stack MERN-Based E-Commerce)
Maximum Marks	4 Marks

Solution Architecture:

Designing a strategic solution architecture is a fundamental step in aligning technical frameworks with overarching business goals. For the ShopEz platform, the architecture is meticulously crafted to support high-traffic product management and execute bulletproof transactions through the MERN ecosystem.

- Utilize the MERN stack to overcome common e-commerce interface and management hurdles.
- Implement a robust data schema for Users, Products, and Orders using MongoDB database solutions.
- Prioritize security in API interactions through Express.js middleware and JWT protocols.
- Guarantee high levels of UI interactivity with a React-based single-page application framework.

Figure 1: High-Level MERN Architecture for ShopEz





Data Layer
(MongoDB Community)

Media Storage
(Product Images)

Presentation Layer (React): Consists of two specialized frontends—one for consumer navigation and another for merchant-level CRUD operations—built using modular UI components.

Server & Interface Layer (Node/Express): Manages standard HTTP requests, performs server-side validation, and directs traffic, ensuring high availability through asynchronous processing.

Persistence Layer (MongoDB): Employs NoSQL collections for dynamic storage of site entities, accompanied by dedicated local environments for handling media assets.

Data Workflow: User events initiate API calls; the server verifies identity via JWT, interacts with the MongoDB cluster, and delivers JSON responses for dynamic frontend rendering.